

Five-year Trends in Various Communicable Diseases Reported under the Integrated Disease Surveillance Program during 2014–2019 in Puducherry

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Abstract

The current study aimed to determine the trend in the incidence of various communicable diseases reported under the Integrated Disease Surveillance Program (IDSP) in Puducherry. This record-based study obtained disease details in terms of the number of suspected cases reported, tested, and positive for the selected diseases reported under IDSP from 2015 to 2019. Dengue disease has shown a recent declining trend in the number of cases and test positivity (7.2% in 2015 to 5.2% in 2019), and there has been a steady increase in the burden of Chikungunya over the years (4.1% in 2015 to 12.6% in 2019). Malaria has been controlled well in Puducherry, but the testing rate was only 3%–4%, well below the national recommendation of testing 10% of the total population. In 2018, there was a recent increase in influenza cases, with test positivity reaching up to 21%. The study unveils valuable insights into communicable diseases in Puducherry, highlighting the fluctuating burden of vector-borne, water-borne, and influenza infections.

Keywords: Communicable diseases, dengue, Integrated Disease Surveillance Program

INTRODUCTION

Worldwide, there has been an epidemiological transition of communicable to non-communicable diseases as the primary cause of morbidity and mortality.^[1] Infectious and associated diseases have reduced by more than half in 2016 from its contribution to 60% of the total disease burden in 1990.^[2] In India, communicable diseases, maternal, perinatal, and nutritional conditions have contributed to one-third of the disease burden in 2016.^[3] Though the burden has reduced over the past two to three decades, certain infectious conditions such as diarrheal diseases, lower respiratory tract infections, and tuberculosis remain the leading cause of morbidity and mortality in India.^[3] Hence, it is essential to regularly monitor infectious diseases to keep them in check and act early in case of any new outbreaks.

In India, the Integrated Disease Surveillance Program (IDSP) conducts regular surveillance of infectious diseases

by establishing an integrated, decentralized, state-based surveillance system. It has an early warning system for timely and effective response to disease outbreaks and health challenges.^[4] It also provides valuable insights into the trends of various diseases. An in-depth understanding of the burden of various infectious diseases and their trends would help make relevant policy decisions. However, there is a lack of published articles on IDSP data from Puducherry since its implementation in the Union Territory. Hence, this study determined the trend in the incidence of various communicable diseases reported under IDSP in Puducherry from 2015 to 2019.

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METHODS

Study design and study setting

This record-based study involved collecting retrospective data from the IDSP web portal for 5 years (2015 to July 2019). Puducherry is a union territory located in the southern part of India. It has four districts: Puducherry, Karaikal, Mahe, and Yanam. The total population of Puducherry district was 9,50,289, according to the 2011 census.^[5]

The health system of Puducherry has an adequate number of health facilities, with good coverage of the population residing in the service area, including migrants. Puducherry IDSP was implemented in Puducherry in 2006. The reporting of the “S form: Syndromic surveillance,” “P form: Presumptive surveillance,” and “L form: Laboratory surveillance” occurs weekly from the subcenter and primary health center. The data is transmitted to the district and the state (UT) surveillance unit and uploaded to the IDSP portal after data verification. L forms are also obtained from tertiary care facilities and medical colleges. The reporting rate was more than 95% in IDSP Puducherry as per the National Center for Disease Control ranking.

Study procedure

The study was done after obtaining permission from the Puducherry Union Territory Health and the institutional ethics committee. All the reported data for the Puducherry district during the last 5-year period (August 2014–July 2019) from the IDSP portal was extracted using a data extraction form. The IDSP epidemiologist trained the data collector (RD) for data extraction. The details of disease in terms of the number of suspected cases reported, tested, and positive were collected for the selected diseases reported under IDSP:

Vector-borne diseases: Malaria, Chikungunya, Dengue

Water and food-borne diseases: Leptospirosis, Shigella, Typhoid

Viral diseases: Hepatitis A and E and influenza

Statistical analysis

Data were entered and analyzed in Microsoft Excel. The number of tests done, positive cases, and outbreaks among the diseases reported in IDSP were summarized as frequency and proportions and graphically represented through a line diagram.

RESULTS

Table 1 shows the percentage change in test positivity for the diseases reported under IDSP between 2015 and 2019 in Puducherry.

Vector-borne diseases

Concerning vector-borne diseases, the burden of Chikungunya and Dengue was high, and the malaria burden was low. Chikungunya had the highest test positivity and second highest number of cases among the diseases reported under IDSP in Puducherry. There has been an increase in test

positivity over the years, from 4.1% in 2015 to 12.6% in 2019. The number of cases has also increased from 7 in 2015 to 404 in 2019, with an outbreak reported in 2018 and 2019 [Figure 1]. Dengue had the second-highest test positivity and highest number of cases among the diseases reported under IDSP in Puducherry. However, there has been a decline in the test positivity rate over the years from 7.2% in 2015 (with spike up to 15% in 2017) to 5.2% in 2019. The number of cases has also declined from 787 in 2015 (with a spike of cases up to 4930 in 2017) to 493 in 2019 [Figure 2]. There has been one outbreak of dengue reported in all the years from 2015 to 2019 except 2018. The burden of malaria has been consistently low over the past 5 years in Puducherry. The number of cases has been less than 20 over all these years, with only two cases in 2019. The test positivity has also been less than 0.05% over all these years, and no outbreak has been noticed.

Water and food-borne diseases

The trend of leptospirosis has also been consistently low between 2015 and 2019 in Puducherry. The number of cases has been less than 10, and test positivity has been less than 1% over the past 5 years, with only a spike during 2017 with 18 cases and test positivity of 1.2%. With Shigella, there has been an alternate increasing and decreasing trend and test positivity over the past 5 years in Puducherry. The maximum number of patients (19) and test positivity (6%) was reported in 2018. Though typhoid cases had spiked from 121 cases in 2015 to 739 cases in 2016, there has been a consistent decline over the next 3 years, with the latest year (2019) having only 41 cases. The test positivity was also maximum during 2016 (3.9%) and least during the year 2019 (0.38%) in Puducherry.

Viral diseases

There has been an alternate increasing and decreasing trend in the burden of Hepatitis A. The maximum burden of cases (21) and test positivity (2.85%) was reported in 2016. The number of cases in the latest year, 2019, was 7, and test positivity was 1%. The burden of hepatitis E has been consistently low over the past 5 years, with only one case reported in the last 2 years (2018 and 2019). Puducherry also had low test positivity (0.2%) over the past 2 years.

The burden of influenza was zero during the initial 2 years, and the number of tests was also lower. However, there has been a spike in the number of cases and test positivity from 2017. The maximum number of cases (319) and highest test positivity (21.17%) were reported in 2018 [Figure 3].

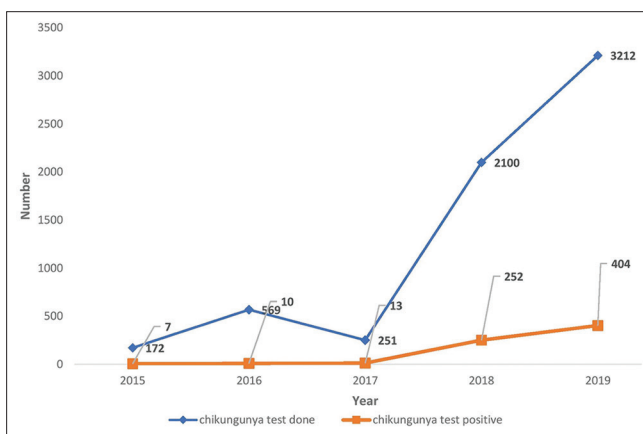
DISCUSSION

IDSP is a critical system that plays a significant role in disease surveillance and outbreak response in all the country's states. Examining the 5-year data of indicator-based reports of all the communicable diseases weekly at IDSP, we found that there has been no specific pattern of vector-borne diseases, water-borne diseases, or influenza in the community of Puducherry.

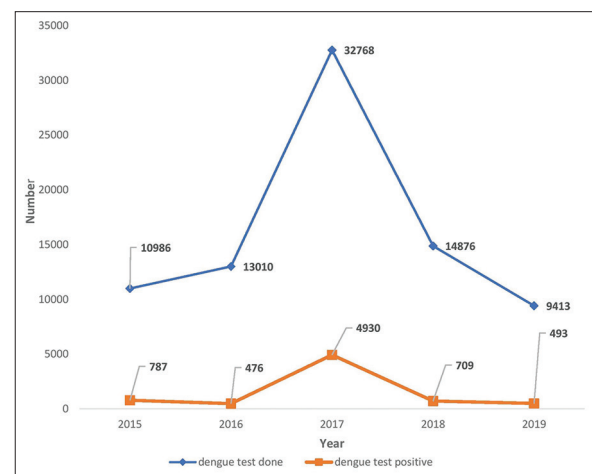
Table 1: Year-wise disease frequency of various diseases reported under IDSP in Puducherry (2015–2019)

Disease	2019		2018		2017		2016		2015	
	Test (n)	Positivity %	Test (n)	Positivity %	Test (n)	Positivity %	Test (n)	Positivity %	Test (n)	Positivity %
Chikungunya										
Test positive	404	12.6	252	12.0	13	5.2	10	1.8	7	4.1
Test done	3212		2100		251		569		172	
Dengue										
Test positive	493	5.2	709	4.8	4930	15.0	476	3.7	787	7.2
Test done	9413		14,876		32,768		13,010		10,986	
Malaria										
Test positive	2	0.006	14	0.030	15	0.030	12	0.031	5	0.012
Test done	32,771		47,061		49,393		39,227		40,575	
Leptospirosis										
Test positive	3	0.3	7	0.5	18	1.2	10	0.9	1	0.1
Test done	900		1312		1550		1117		1231	
Shigella										
Test positive	18	5.0	19	6.0	9	2.6	13	2.8	14	2.8
Test done	357		317		352		468		500	
Typhoid										
Test positive	41	0.38	115	0.69	153	0.62	739	3.9	121	1.18
Test done	10,749		16,784		24,567		18,972		10,272	
Hepatitis A										
Test positive	7	1.0	3	0.41	12	1.12	21	2.85	7	1.38
Test done	697		730		1073		737		506	
Hepatitis-E										
Test positive	1	0.2	1	0.2	11	1.8	11	1.8	0	0
Test done	484		592		614		621		398	
Influenza										
Test positive	26	1.99	319	21.17	142	19.26	0	0	0	0
Test done	1304		1507		737		1		58	

¹Positivity (%) total number of test positive among the sample taken for the concerned disease

**Figure 1: Time trend of chikungunya reported to IDSP during 2015–2019**

Vector-borne diseases such as dengue and chikungunya were found to have the highest burden of cases and test positivity in Puducherry. However, dengue has shown a recent declining trend in the number of instances and test positivity (7.2% in 2015 to 5.2% in 2019), while there has been a steady increase in the burden of chikungunya over the years (4.1% in 2015 to 12.6% in 2019). Such differing trends of these two diseases

**Figure 2: Time trend of dengue reported to IDSP during 2015–2019**

were surprising, given that both are similar in epidemiology, transmission cycle, vector (*Aedes* species), and clinical symptoms during onset.^[6] However, there is a possibility of re-emergence of dengue (as seen in 2017 with test positivity of about 15%) given Puducherry's ever-growing urbanization, climatic, ecological, and environmental factors. The emergence

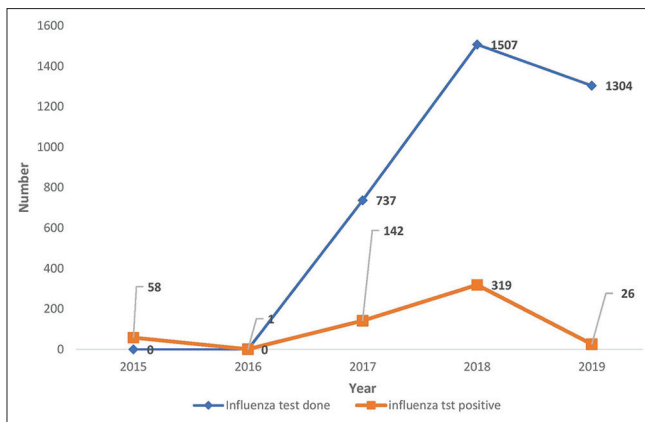


Figure 3: Trends of influenza reported to IDSP during 2015–2019

of chikungunya and the re-emergence of dengue should be dealt with through vector control measures as no specific line of management is available for both conditions.^[7]

Malaria has been controlled well in Puducherry and is aligned to the National Framework for Malaria Elimination Targets.^[8] It is noteworthy that malaria testing has been consistently high, irrespective of the decrease in the disease load. However, the testing rate needs to be further improved from the current 3%–4% rate to the national recommendation of testing 10% of the total population in Puducherry.^[9]

All the water and food-borne diseases such as leptospirosis, shigella, typhoid, and viral diseases with the fecal-oral route of transmissions such as Hepatitis A and E has been kept well under control over the past 5 years in Puducherry. Such a consistently lower trend can be attributed to Puducherry's safe water supply and good sanitation practices. The water distribution system has been improved significantly with a coordinated response from the community, political commitment, and regulatory action. Clean and safe sanitation practices are another aspect that has been improved over the years in Puducherry. The lower burden of water and food-borne diseases indirectly reflects improvement in these aspects. Furthermore, the ongoing social development might also affect the recent burden of the disease and was not assessed in the current study.

We also noticed a recent spike in influenza cases, with test positivity up to 21% during 2018 in Puducherry. Such a trend is worrying as we are already facing a COVID-19 pandemic, and a simultaneous rise in the burden of influenza challenges the healthcare system in Puducherry. Currently, the focus is highly on the testing of COVID-19, and influenza testing might be neglected. There has also been evidence saying that the non-pharmacological interventions (NPIs) taken for the COVID-19 pandemic, such as hand hygiene, social distancing, movement restrictions, closure of schools, colleges, and workplaces, and limiting the number of people during gatherings might help in reducing the burden of influenza also.^[10] However, previous pandemic experiences have reported a limited role of these NPIs, given the differences in influenza's

characteristics and transmission dynamics.^[11,12] Hence, the testing and control of influenza should also be considered during this testing phase of the COVID-19 pandemic.

The study's strengths are that we also reported the test positivity and the burden of cases to gather more reliable evidence on the epidemiology of these conditions. We have also used data from the entire Puducherry, making the study results generalizable. As the data were retrieved from the IDSP portal, data duplication and manipulation were impossible, and appropriate data quality checks were made before entering the portal. However, our study has certain limitations. We have not considered age- or gender-specific distribution of the conditions as it will give a better picture of the burden and develop measures targeting only a particular population. The changes in the disease trend after the COVID-19 pandemic could not be studied. Future studies can focus on having the individual as a study unit to explore the possible reasons for the emergence, re-emergence, or outbreaks of any conditions reported under IDSP in Puducherry.

However, this study provides important information regarding the pattern and frequency of disease and the availability of resources and infrastructure in the healthcare system of Puducherry. The data from the surveillance system were used to evaluate the impact of infection and prevention control programs along with indicator-based achievement and progress of eradication and elimination of various communicable diseases. We need to control the epidemic and outbreak of diseases through various health system strengthening measures, capacity building, training, and optimal use of resources. Surveillance raises the faith and bonding of public health workers, community leaders, and their volunteer activity. Improving surveillance strategy or mixing evidence and indicator-based situational standard operating procedures can better respond to health emergencies. A parallel non-communicable disease surveillance system can be considered with an objective primarily to focus on early diagnosis of conditions and preventing diseases by lifestyle and behavioral modifications.

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Conflicts of interest

There are no conflicts of interest.

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